1 A load is suspended from a thread. The vertical force on the thread due to the load is 0.75 N.

(a) Calculate the mass of the load.

mass = ... [2]

(b) Fig. 1.1 shows the load suspended from the thread.

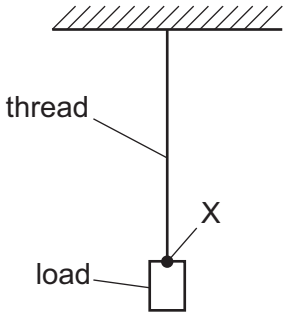

Fig. 1.1

A wire is attached to the load at point X and pulled horizontally to the right.

The tension in the horizontal wire is 1.2 N.

By drawing a scale diagram or by calculation, determine:

- the magnitude of the resultant of the force at X due to the load and due to the tension in the wire
- the direction of the resultant relative to the vertical direction.

Show your working.

magnitude of resultant force = ... N

direction of resultant relative to vertical = ... °

[4]

DO NOT WRITE IN THIS MARGIN

(c) Forces may produce changes in the size and the shape of an object.

State **two** other changes that forces may produce.

1 .....

2 .....

[2]

[Total: 8]

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